

Cluster Structure Revealed by the GDR-4 Dimensionality Reduction Method in Anthropometric Data

Roman Kotlowski

09.07.2026

Abstract

This paper presents an analysis of clusters obtained using the GDR-4 dimensionality reduction method applied to a 54-dimensional anthropometric dataset. The reduction process followed a multi-stage projection scheme:

$$54 \rightarrow 36 \rightarrow 24 \rightarrow 16 \rightarrow 8 + 4 \rightarrow 8 \rightarrow 4 + 2 \rightarrow 4,$$

which preserves local geometric relationships and proportional structure across semantic blocks of anthropometric features. Clustering was performed using DBSCAN with parameter $\varepsilon = 14$, resulting in four multi-element clusters and five singleton clusters. The results demonstrate that GDR-4 maintains both global and local structure of high-dimensional anthropometric data.

1 Introduction

High-dimensional anthropometric datasets contain strong internal correlations and block structures, such as limb lengths, circumferences, anatomical heights, and head dimensions. Classical dimensionality reduction methods, such as Principal Component Analysis (PCA), often distort these structures by mixing features into global linear combinations.

The GDR-4 method was designed as a hierarchical, block-preserving projection technique. Its goal is to reduce dimensionality while maintaining interpretability and proportional relationships between anthropometric features. The final 4-dimensional space retains meaningful geometric information from the original 54-dimensional dataset.

2 Methodology

2.1 GDR-4 Dimensionality Reduction

The reduction pipeline consists of several projection stages. Each stage removes redundancy within semantic blocks while preserving dominant geometric directions. The final

4-dimensional representation is stable, interpretable, and suitable for clustering.

2.2 Clustering Procedure

DBSCAN was applied to the 4-dimensional GDR-4 output with $\varepsilon = 14$. This small value of ε allows the detection of fine-grained local structures, including micro-clusters and singleton profiles.

3 Results

The clustering produced four main clusters and five singleton clusters. Figure 1 shows the cluster structure obtained from the GDR-4 representation.

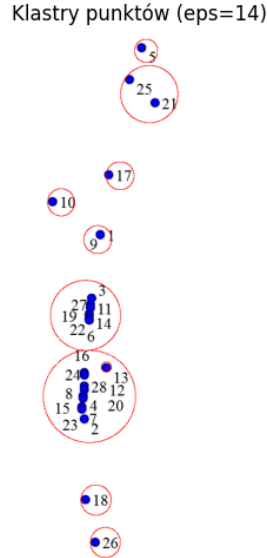


Figure 1: Clusters obtained from the 4-dimensional GDR-4 representation using DBSCAN with $\varepsilon = 14$.

3.1 Cluster 1: {21, 25}

A two-element cluster representing nearly identical anthropometric profiles. These individuals exhibit minimal differences across multiple blocks of features, including limb lengths, circumferences, and anatomical heights.

3.2 Cluster 2: {3, 6, 11, 14, 19, 22, 27}

A seven-element cluster representing individuals with balanced and consistent body proportions. Their limb lengths, circumferences, and anatomical heights form a coherent geometric group in the reduced space.

3.3 Cluster 3: {2, 4, 7, 8, 12, 13, 15, 16, 20, 23, 24, 28}

The largest cluster, containing twelve individuals. This group represents the “central mass” of the dataset: profiles with typical, non-extreme anthropometric characteristics. Their proportions are close to the population average.

3.4 Cluster 4: {9, 1}

A second two-element cluster of highly similar individuals. Their anthropometric profiles differ only slightly across multiple feature blocks, forming a compact pair.

3.5 Singletons: {5, 10, 17, 18, 26}

Five individuals form singleton clusters. These profiles are not global outliers but exhibit locally unique combinations of anthropometric features. Such uniqueness may arise from atypical limb proportions, unusual circumferences, or distinctive anatomical height patterns.

4 Discussion

The results confirm that GDR-4 preserves both global and local structure of anthropometric data. The presence of micro-clusters and singleton profiles demonstrates high local resolution in the reduced space. Unlike PCA, which tends to blur fine-grained structure, GDR-4 maintains proportional relationships between feature blocks.

The largest cluster corresponds to typical anthropometric profiles, while the smaller clusters represent groups with strong internal similarity. Singletons highlight individuals with unique geometric configurations.

5 Conclusion

The GDR-4 method effectively reduces high-dimensional anthropometric data while preserving meaningful geometric structure. Clustering with DBSCAN at $\varepsilon = 14$ reveals four coherent clusters and five unique profiles. These findings demonstrate the suitability of GDR-4 for applications requiring interpretability, structural preservation, and high-resolution local analysis.